



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	ANALYTICAL TECHNIQUES IN WATER QUALITY ASSESSMENT				
Type of Course	SEC				
Semester	VI				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	2	-	1	45
Pre-requisites	1. Basic idea on volumetric Analysis 2. Knowledge on Water distribution and water resources				
Course Summary	1. To enable the students to become aware of the water quality standards and to familiarize the methods for analysing water qualities. 2. To make them aware of the impact of water pollution and hence reduce water pollution.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the analytical techniques used in chemistry for water quality monitoring	U	C	Instructor-created exams / problem solving
CO2	Demonstrate the instrumental methods used in water quality monitoring	U	P	Instructor- Created exams/assignment
CO3	Enhancing the knowledge on water purification methods	U	C	Presentation- Peer Teaching
CO4	Demonstrate the procedures for the determination of water quality	Ap	P	Problem solving- Home Assignments
CO5	Acquire skill on the analytical techniques used in water Analysis	Ap	P	Doing practicals
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Analytical techniques in Chemistry		8	15
	1	An introduction to analytical methods in chemistry- concentration terms- Molarity, Molality, Normality, v/v, w/v, ppm and ppb, Dilution of solutions, standard solutions	4	
	2	Principles of Volumetric methods in water Analysis- Acid-base titrations, Redox titrations,	2	
	3	complexometric titration and precipitation titrations	2	
II	Instrumental Methods of Water Analysis		9	18
	4	pH meter, Conductivity meter, Turbidity meter, Flame photometer, Colourimeter,	2	
	5	Atomic absorption spectrophotometer (AAS), GCMS	2	
	6	Ion-selective electrodes, Isotopic analysis	1	
III	Water Purification		9	18
	7	Water resources: ground water and surface water, Importance of water, Water quality- water cycle,	1	
	8	Distribution of water -Water scarcity, Common water quality problems	1	
	9	Potable Water: Pre-treatment, coagulation, filtration, disinfection,	2	
	10	Water storage supply, Demineralization & desalination		
	11	Water softening methods-Lime soda process- Zeolite process, Ion exchange method	2	
	12	Sewage waste water treatment-Primary, Secondary and tertiary treatment & sewage water treatment plants in Kerala	2	
	13	Desalination of brackish water-Electrodialysis-Reverse Osmosis	1	
IV	Determination of Water quality parameters		10	19
	14	Water quality monitoring	1	
	15	Water Quality parameters- Odour, Temperature, Colour, Turbidity, pH, Total Dissolved Solids (TDS), Conductivity-	2	
	16	Experimental methods for the estimation of Alakalinity, Hardness of water	2	
	17	Estimation of anions and cations from dissolved minerals	1	
	18	Experimental methods for the estimation of Biological parameters- DO, BOD, COD, Microbiological parameters,	2	
	19	Water quality standards: drinking water- WHO guidelines- Water quality standards by BIS, IS-10500-2012 on drinking water specification,17482-2020-on drinking water supply management system	2	
IV	Water Quality Assessment (Open Ended)		9	
	The following water quality assessment may be through hands on training			

	1	Determination of -temperature, pH, conductivity-Instrumental Methods		
	2	Determination of turbidity using Turbidity meter		
	3	Determination of total dissolved solids (TDS)-Gravimetric/Instrumental method		
	4	Determination of carbonate and bicarbonate using titration method (Acidimetry & Alkalimetry),		
	5	Determination of Ca, Mg, Total hardness -Complexometry		
	6	Determination of Ammonia and iron-Colorimetry		
	7	Determination of Dissolved Oxygen (DO)- Winkler's Method-Iodometry		
	8	Determination of Chloride- Argentometry		

References:

1. R. Ramesh & M. Anbu, Chemical methods for Environmental Analysis: Water and Sediment, Madras Macmillan.
2. B.K. Sharma, Instrumental methods of chemical analysis, Krishna Publication Media (P) Ltd. Meerut.
3. S.S. Dara, A Textbook of Environmental Chemistry and Pollution Control, 8th Edition, S. Chand and Sons, New Delhi
4. B.K. Sharma and H. Kaur, Environmental Chemistry, Goel Publishing House, Meerut
5. Water Pollution- causes, effects and control- P K Goel, New age International

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	-	-	-	-	-						
CO 2	2	3	-	-	-	-						
CO 3	-	-	1	-	-	-						
CO 4	-	-	2	3	-	-						
CO 5	-	1	-	-	-	-						

CO 6	-	-	-	3	-	-						
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Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Assignments(20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1		✓		✓
CO 2	✓			✓
CO 3	✓			✓
CO 4		✓		✓
CO 5			✓	✓